

2021 Ph H2 Q15

Section: Electricity

Topic: Semiconductors, Sources, Internal Resistance

This question investigates the internal resistance of a cell.

(a)

Calculate the current in the circuit

Given:

- $V_{\text{external}} = 2.4 \text{ V}$ across the 6.0Ω resistor
- Use Ohm's Law:

$$I = \frac{V}{R} = \frac{2.4}{6.0} = 0.40 \text{ A}$$

(b)

Calculate the internal resistance of the cell

Use the formula for terminal voltage:

$$V = \mathcal{E} - Ir \quad \Rightarrow \quad r = \frac{\mathcal{E} - V}{I}$$

Given:

- $\mathcal{E} = 3.0 \text{ V}$
- $V = 2.4 \text{ V}$
- $I = 0.40 \text{ A}$

$$r = \frac{3.0 - 2.4}{0.40} = \frac{0.6}{0.40} = 1.5 \Omega$$

Revision Tips

- Terminal voltage is the voltage measured **across the external resistor**, not the full EMF
- Internal resistance causes a **voltage drop** inside the source:

$$V = \mathcal{E} - Ir$$

- The bigger the current, the bigger the drop across internal resistance
- Always use **consistent units** — volts, ohms, amperes